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Projectile loader having streamlined external body and internal opening mechanism

Abstract

A projectile loader is provided having a body split into two portions attached by a hinge, wherein a release mechanism for separating the two portions is located internally of the loader body, accessed via a top opening.

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Background/Summary

RELATED APPLICATION DATA (1) The present application is a Continuation of U.S. Non-Provisional patent application Ser. No. 17/104,533, filed Nov. 10, 2020, which claims the benefit of U.S. Provisional Patent Application No. 62/940,676, filed Nov. 26, 2019, the entirety of each of which are incorporated by reference as if fully set forth herein.

FIELD OF THE INVENTION

(1) This invention relates to the field of projectile loaders, and more particularly, to a projectile loader having an internal mechanism for releasing a hinged loader body for a streamlined external appearance.

BACKGROUND

(2) Projectile loaders (otherwise known and used interchangeably herein as hoppers, magazines, projectile loaders, or loaders) generally connect to projectile markers (compressed gas guns) and feed projectiles into the marker. These projectile loaders store projectiles and have an outlet or exit tube (outfeed tube or neck). The outlet tube is connected to an inlet tube (or feed neck) of the projectile marker, which is in communication with the breech of the marker. During the normal operation, projectiles dropped through the outlet of the loader from a projectile stack within the outlet tube and gun inlet tube. When the projectile at the bottom of the stack is dropped into the firing chamber of the projectile marker, it is replaced at the top of the stack from the supply of projectiles remaining in the loader housing, thereby replenishing the stack. Thus, the loaders act to hold and feed projectile projectiles into the breach of a projectile marker, so that the projectiles can be fired from the marker.

(3) Many loaders contain agitators or feed systems to mix, propel, or otherwise move projectiles in the loader. This mixing is performed by an impeller, projection, drive cone, agitator, paddle, arm, fin, carrier, or any other suitable mechanism, such as those shown and described in U.S. patent application Ser. No. 12/171,956 (U.S. Patent Application Publication No. 2009/0133680) and U.S. Pat. Nos. 6,213,110; 6,502,567; 5,947,100; 5,791,325; 5,954,042; 6,109,252; 6,889,680; and 6,792,933; the entire contents of all of which are incorporated by reference herein as if fully set forth. In “gravity feed” hoppers, the hopper has no moving parts, and projectiles fall into an outfeed tube by gravity. In “agitating” loaders, an agitator mixes projectiles so that no jams occur at the exit opening of the outlet tube. In “force feed” or “active feed” projectile loaders, the agitator or feeder (drive cone, carrier, paddle, or any other force feed drive system) forces projectiles through the exit tube. As the firing rates of projectile markers have increased, active feed loaders have become extremely popular due to their ability to ensure the proper feeding of projectiles and sustain the feed rates necessary to support today's markers.

(4) Modern loaders utilize projections, paddles, arms, carriers, drive cones, or other agitators to mix or advance projectiles. These agitators are operated by motors which are often controlled by an electronic control circuit. These control circuits may utilize microprocessors in conjunction with at least one sensor configured to detect any number of parameters, including the firing of a projectile from the marker or the occurrence of a jam. These components are typically powered by a D.C. power source, such as a battery.

(5) It is often necessary for, by way of example, a projectile sport player to access the interior components of a projectile loader, for example to fix a jam. A player may also need to access components inside the body of the projectile loader, such as the agitators, control circuits, power supplies, sensors, and drive mechanisms.

(6) Moreover, if a projectile breaks inside a projectile loader, there must be a convenient and efficient way to open the body and clean the paint or dye from the inside of the projectile hopper, without having to fully disassemble the loader.

(7) Virtually all of the known solutions to the issue of accessing a projectile loader have focused on typical external release mechanisms such as push-buttons or latches. These have the drawback of being prone to damage, dirt, and debris, along with the possibility of accidental actuation during activity. Likewise, such external components add visual weight that detracts from the sleek aesthetic that players desire.

(8) Thus, there is the need for an openable projectile loader and feed mechanism that overcomes the aforementioned problems and provides easy release/open of the loader while maintaining a uniform outer appearance.

SUMMARY

(9) The present invention provides for an openable projectile loader comprising a body defining an interior area housing internal components and an internally accessible release mechanism. The body has a generally smooth outer surface, with few or no external protrusions, recesses or features. The body is split into a first portion and a second portion, the first portion is attached to

the second portion by a hinge, and the first portion and second portion are moveable relative to each other via the hinge between an open position and a closed position. The release mechanism is configured to selectively permit detaching the first portion and the second portion, wherein the release mechanism is located in the interior area of the body.

(10) The projectile loader has an opening for stocking the projectile loader, wherein the release mechanism is accessible via the opening.

(11) In another embodiment, the actuator for the release mechanism can be located proximate to the front of the projectile loader and the opening being located proximate to the rear of the projectile loader.

(12) In another embodiment, the release mechanism actuator can be located proximate to the hinge.

(13) In another embodiment, the first portion and the second portion of the projectile loader can be essentially the same size.

(14) In another embodiment, the present invention provides a moveable feeder housed within the interior of the projectile loader body.

(15) In another embodiment, the present invention provides a feed mechanism comprising a moveable feeder and a motor housed within the interior of the projectile loader body, and the feed mechanism is configured to be at least partially removable from the projectile loader body when body is in the open opposition.

(16) In another embodiment, the first portion and the second portion of the projectile loader are removably engaged.

(17) In another embodiment, the present invention provides the hinge being spring-biased to an open position.

(18) In another embodiment, the present invention provides the hinge being spring-biased to a closed position.

(19) In another embodiment, the present invention provides a projectile loader with the body being split generally vertically along the length of the body such that the first and second portions are respectively a first side portion and a second side portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a side elevational view of an exemplary projectile loader attached to a projectile marker (gun) shown in broken lines.

(2) FIG. 2 is a side partial cutaway of a projectile loader, showing the internal feed mechanism, with a portion of the body shown in phantom.

(3) FIG. 3 is a side perspective view of a projectile loader having a hinged body according to an embodiment of the present invention.

(4) FIG. 4 is a rear perspective view of the projectile loader according to FIG. 3.

(5) FIG. 5 is a side cross-sectional view of the projectile loader according to FIG. 3.

(6) FIG. 6 is a partial cutaway of the projectile loader according to FIG. 3.

(7) FIG. 7 is a partial side cross-sectional view of the projectile loader according to FIG. 3.

(8) FIG. 8 is a partial top view of a bottom of the projectile loader of FIG. 3 with a first portion of the loader and a feed mechanism both removed.

(9) FIG. 9 is a side view of another aspect of a projectile loader having a hinged body according to another embodiment of the present invention.

(10) FIG. 10 is a side perspective view of the projectile loader according to FIG. 9 in an open position with the top lifted to expose the internal release mechanism.

(11) FIG. 11 is a partial top view of the projectile loader according to FIG. 9.

(12) FIG. 12 is another partial top view of the projectile loader according to FIG. 9.

(13) FIG. 13 is a side perspective view of the projectile loader according to FIG. 9, with a portion of the loader removed to show component parts of a release mechanism.

(14) FIG. 14 is an exploded view of component parts of another feed mechanism for use with a hinged projectile loader body of the present invention.

DETAILED DESCRIPTION

(15) FIG. 1 is a side elevational view of a loader **100** or **900** operatively attached to a representative projectile marker (gun) **20**, illustrated in broken lines. The projectile marker **20** includes a main body **22**, a compressed gas cylinder **24**, a front handgrip **26**, a barrel **28**, and a rear handgrip **30**. The projectile marker **20** also includes an inlet tube **32** leading to a firing chamber in the interior of the main body and a trigger **34**. The front handgrip **26** projects downwardly from the barrel **28** and provides an area for gripping by an operator of the projectile marker **20**. The compressed gas cylinder **24** is typically secured to a rear portion of the projectile marker **20**. The compressed gas cylinder **24** normally contains CO₂, although any compressible gas may be used.

(16) A projectile loader or "loader" **100/900**, such as one which may be used for the present invention, includes a body **101/901** which houses an open interior area **138/938**, a rear end **134/934**, and a front end or front portion **132/932**. The loader **100/900** may include a portion **42** having control circuitry or electronics, which may include a microprocessor, for controlling various operating parameters of the projectile loader. The lower portion of the projectile loader leads to an outfeed, exit tube, or feed neck **112/912**, which communicates with the infeed tube **32** to supply projectiles to the projectile marker **20**. A manual drive assistance actuator **60** may be provided to turn the feeder inside the projectile loader body **101/901**.

(17) In operating the projectile marker **20**, the trigger **34** is squeezed, thereby actuating the compressed gas cylinder **24** to release bursts of compressed gas. The bursts of gas are used to fire projectiles outwardly through the barrel **28**. The projectiles are continually fed by the loader **100/900** through the inlet tube **32** to the firing chamber.

(18) Some projectile loaders or feeders are described in detail in U.S. Pat. No. 6,213,110 ("Rapid Feed Projectile Loader"), U.S. Pat. No. 6,502,567 ("Rapid Feed Projectile Loader With Pivotal Deflector"), U.S. Pat. No. 6,701,907 ("Spring Loaded Feed Mechanism For Projectile Loader"), and U.S. Pat. No. 6,792,933 ("Drive Cone For Projectile Loader"), the entire contents of which are each incorporated by reference herein as if fully set forth.

(19) As shown in FIG. 2, a feed mechanism **56** is generally used to drive, feed, move, or urge the projectiles toward the exit tube **52** and into the inlet tube **32** of the projectile marker **20**. The feed mechanism **56** preferably includes a feeder **66** positioned within the projectile loader body and coupled to a drive shaft, with the drive shaft **64** coupled to a motor **58** and gear system **72, 74**, to drive projectiles **68** toward the exit tube **52**. While an illustrative feed mechanism **66** is shown, various other components may be substituted for driving projectiles into the projectile marker **20**. A catch cup portion **70** may house the feeder **66**, and act as a reservoir or well where the projectiles **68** are held for feeding.

(20) A variety of feed mechanisms **56** and feeders **66** can be used in the present invention, including an impeller, drive cone, paddle wheel, fin, carrier, or other device which can direct or otherwise force or urge projectiles into the exit tube **52**. By way of example and not limitation, a drive cone may be used with a plurality of fins which extend in a radial direction from the central part of the drive cone. It is appreciated that the feeder may include recesses or pockets within which the projectiles sit as they are shuttled toward the exit tube **52**. The feeder **66** may be mounted on a drive shaft **64**. The drive shaft **64** is connected to the motor **58** to rotate about a central axis. As the motor **58** operates, the feeder **66** moves, pushing balls into exit tube **52**.

(21) While the feed mechanism **56** is shown, it is appreciated that a projectile loader **100** according to the present invention could be formed without an interior moving feed mechanism or feeder, and work as a gravity fed hopper.

(22) FIGS. 3-8 show an embodiment of a loader **100** according to the present invention, which

includes a body **101** including includes at least a first portion **102** and a second portion **103** with an interior area **138** therebetween. The interior area **138** houses internal components of the loader **100**, such as, by way of example, the feed mechanism **56**, power supply, or source (e.g., batteries or battery pack), and/or circuitry. As shown in FIG. **8**, a plurality of retention elements **111**, for example four retention elements, are provided to hold the feed mechanism **56** within the second portion **103** of the loader body, or to otherwise secure or hold components, or permit relative movement of components.

(23) In one aspect, the first portion **102** and the second portion **103** are essentially the same size. As used in this respect, the term “essentially the same size” with respect to the first and second portions means that these two components each define a partially enclosed volume that is within 20-50% of each other.

(24) The body **101**, as assembled, preferably has a generally smooth outer surface, or an outer surface having few surface elements or features. For example, a loader body **101** according to the invention may preferably have an outer surface or wall that is free from external features other than a single recess for a power or operation button. This design provides a “sleeker” or “streamlined” silhouette than known designs, with less outer surface features. In addition, a release mechanism, described in greater detail below, is positioned on an inner portion of the body, so that is accessible in an interior of the body. Therefore, the release mechanism is not found on the outer surface of the loader body **101**.

(25) Looking directly at the loader **100** from the front or nose portion **132**, the first portion **102** may be all or part of a top of the body **101**, and the second portion **103** may be all or part of a bottom **136** of the body **101**. In the illustrated embodiments, the body **101** is split essentially horizontally along a horizontal plane running generally along the middle of the body **101**, and may have an opening at the bottom **136**. In this manner, the body **101** opens at the bottom **136** to receive interior components of the projectile loader. In some embodiments, the body **101** is split longitudinally along the length of the body **101**, from the nose portion **132** to the back area **134**. As such, it is preferable that no split in the body is provided transversely, such as across the width of the body **101**. In other embodiments, a split in the body may be provided along a vertical plane, such as in a left/right split arrangement, or the split can follow other paths to separate the body **101**.

(26) As shown in FIGS. **5** and **6**, a latch **116** on the second portion **103** engages a catch **117** on the first portion **102** to hold the loader body **101** closed. An internal release mechanism **140** actuates the latch **116** and the catch **117** that the body can be opened, and may operate as an attachment and release mechanism. The internal release mechanism **140** is positioned within an interior or adjacent an inner surface of the body. In a preferable embodiment, the release mechanism **140** does not have any portion on the outer surface of the body. The release mechanism **140** is preferably accessed by a user reaching through the opening **109**. Generally, a “release mechanism” **140** applicable to the present invention provides a manual actuation to separate the first and second portions **102**, **103**, and may include various structures that are operable with one hand. The release mechanism **140** may alternatively provide a button, switch, or other device to activate an electromechanical device, a solenoid, or the like for releasing the latch **116** from the catch **117**. The release mechanism **140** should preferably be positioned so as to be easily used and accessed without contributing to projectile jams or significantly decreasing the volume of the loader. The release mechanism **140** is preferably positioned completely inside the body **101** so that the body retains its streamlined appearance and so as to prevent accidental actuation.

(27) In the illustrated embodiment, the release mechanism **140** includes the latch **116** mounted on the second portion **103** and the latch **117** mounted on the first portion **102**, although the latch and catch may be reversed.

(28) The latch **116** includes a member **146**, illustrated here as a horizontally extending bar, for receiving a portion of the catch **117**. The latch further includes two legs **148**, **150** and a lower member **147**. The legs **148**, **150** extend away from the second portion **103** and create a space

between the member **146** and the back end **134** of the body **101**. The latch **116** may be a rigid structure and may be integrally formed with the second portion **103**.

(29) The catch **117** is pivotable relative to the first portion **102**, by way of flexible or resilient material construction such as plastic, by way of a hinge, or by similar mechanical structure. The catch **117** includes an actuator **142** that may be manipulated to move the catch. As best shown in FIG. 6, the actuator **142** includes a rib **143** and a distal end **144** that are sized and configured to engage the member **146** of the latch **116**. As shown in FIG. 7, the distal end **144** and the member **146** can have complementary angled surfaces that aid in closing the loader body **101**, particularly by partially pressing the actuator as the angled surfaces slide along one another until the distal end **144** is completely below the member **146**.

(30) The actuator **142** may be resiliently biased toward the front portion **132** of the body **101** and the various corners and surfaces of the rib **143** and distal end **144** may be beveled so that the actuator automatically engages the latch **116** when the first portion **102** is closed. To release the first portion, the actuator **142** is moved (i.e., pressed or depressed) toward the rear end **134**. Note that the release mechanism **140** may have a push or pull operation. On an opposite side of the catch **117** from the actuator **142** is an extension **145** sized to fit within a recess **135** on the rear end **134** of the loader body **101**. When the actuator **142** is depressed, the extension **145** fits within the recess **135** and the recess thereby limits how far the actuator **142** can move.

(31) Depressing, pulling, or otherwise actuating the release mechanism **140** releases the body portions **102**, **103** so that they could be separated as shown by the arrow A. The body portions **102**, **103** may remain connected by the hinge **105** at the front **132** of the body **101**.

(32) In the illustrated embodiment, the release mechanism **140** is positioned only on and proximate the rear end **134** of the loader body **101**. Alternatively, the release mechanism **140** may be positioned along the sides or front of the body **101** or portions(s) thereof.

(33) Because the player may not be able to see the release mechanism **140** while operating it, various ergonomic features may be included. In the illustrated embodiment, the release mechanism **140** includes a relatively large and easy to find and utilize actuator **142** that is slightly recessed from the rest of the catch **117**. The release mechanism **140**, particularly the actuator **142**, may also include other ergonomic/detectable features such as a textured surface, a raised rib, a dimple, or the like to aid operation. The illustrated actuator **142** is positioned proximate the opening **109** and the rear portion **134**, which allows a user to grab the loader body **101** with some fingers pressing on the rear portion **134** from the outside to provide leverage while one or more fingers (such as a thumb) press the actuator **142**, such that the user's hand is effectively making a "squeezing" motion.

(34) In use, pressing the actuator **142** of the release mechanism **140** causes pivoting motion of the catch **117** toward the rear end **134**, which moves the rib **143** and the distal end **144** of the catch **117** laterally/horizontally away from the latch **116**.

(35) As a result, the catch **117** is disengaged from the latch **116** and the first portion **102** is free to separate from the second portion **103**. Alternatively, the release mechanism can actuate a solenoid to release the latch **116** from the catch **117**.

(36) The release mechanism **140** may have various other additions or modifications. For example, the release mechanism **140** may provide an audible "click" or similar feedback when successfully actuated for release and/or provide an audible "click" when fully closed. The release mechanism **140** may require a two-step actuation motion, or the actuation of two separate buttons, so as to prevent accidental actuation. A spring, such as a torsion spring or a compression spring, may bias the upper body portion **102** to either the open position or the closed position, as desired.

(37) In one embodiment, a torsion spring is mounted at coaxially with the hinge **105** to both the first and second portions **102**, **103** to bias them toward the open position.

(38) In alternative embodiments, a part of the release mechanism **140** could be configured to be pulled forward toward the front portion **132** to separate the latch **116** from the catch **117**. For example, the rib **143** of the catch **117** or the member **146** of the latch **116** could be hinged or

resiliently cantilevered to allow such movement, which in turn would allow the loader **100** to open. (39) The opening **109** of the loader **100** may be a “speed feed” reloader provided with a plurality of fingers **115** extending from the perimeter of the opening to the center of the opening. The fingers **115** may be formed of a resilient material or otherwise spring-mounted, allowing the fingers to deflect when a user reaches and/or loads projectiles and then return to a position extending across the opening **109**. In this manner, the fingers **115** function to retain projectiles in the loader **100**. (40) As shown in FIGS. **9** and **10**, another embodiment of an openable projectile loader **900** according to the present invention is shown with like numerals representing similar parts. The loader **900** includes a body **901** with an interior area **938**, which houses internal components as discussed above.

(41) The body **901** includes at least a first portion **902** and a second portion **903** that are split longitudinally. Each of the first portion **902** and second portion **903** include a front or nose portion **932**, top wall **908**, an opening **909** or mouth area for receiving projectiles, a rear or back area **934**, a lower rear wall **911**, a feed neck portion **912**, and a battery area **913**, and a lower front wall **914**. The opening **909** may be covered with a lid **915** that is removable, hinged, or otherwise temporarily attached to the body **901**. As in the previous embodiment, the first portion **902** and the second portion **903** may be split in various ways and shapes, and may be in varying relative proportions of the overall loader body **901**. For example, the second portion **903** may be the majority of the projectile loader body, and the first portion **902** may be a smaller panel that opens, or vice versa. The body **901** has an opening at the bottom **936** to receive interior components of the projectile loader.

(42) The first portion **902** and the second portion **903** are attached by a hinge **905**. The hinge is positioned along a division seam **906** between the first portion **902** and the second portion **903** and may be either external (as illustrated) or internal to the loader body **901**. The hinge **905** may be any type of hinge, including, for example, but not limited to, a mechanical hinge of any kind, a hinge such as on a door or cabinet, a butt hinge, living hinge, or strap hinge. One or more hinges may be provided. The hinge may be self-closing such as with a spring-biased hinge, or otherwise biased to a closed or open position. The first portion **902** and the second portion **903** move relative to each other as indicated by arrow A in FIG. **13**, via the hinge **905**. Although the hinge **905** is illustrated proximate a front portion **932** of the body **901**, the hinge **905** may alternatively be positioned anywhere along the division seam **906**. Thus, a moveable joint or mechanism is provided to allow movement of the first portion **902** and the second portion **903**.

(43) FIGS. **9-13** depict a projectile loader body **901** according to the present invention in use with the internal components of a well-known projectile loader, offered under the EMPIRE® and HALO® brand names. The feed mechanism may be similar to that shown in FIG. **2**. The body **901** comprises a first portion **902** and a second portion **903**, attached by a hinge **905**. The hinge **905** is located along the division seam **906** along the front portion **932** of the body **901**. As shown in FIG. **6**, the first portion **902** and second portion **903** each comprise about half of the projectile loader body, although other relative portions can be used when dividing the body.

(44) As shown in FIGS. **10** and **12**, a latch **916** on the second portion **903** engages a catch **917** on the first portion **902**. The latch **916** and catch **917** are manipulated and actuated by a release mechanism **940** that is positioned in a portion of the body to release the two body portions so that the body can be opened, and may operate as an attachment and release mechanism. Generally, a “release mechanism” **940** applicable to the present invention provides a manual actuation to separate the first and second portions **902**, **903**, and may include, for example, a mechanical linkage, a bowden cable **210** (as further discussed below), an electromechanical device, a solenoid, or the like. As with the embodiment of FIGS. **3-8**, the release mechanism **940** is preferably accessed by a user reaching through the opening **909** when the lid **915** is removed/open. The release mechanism **940** should be positioned so as to be easily used without contributing to jams or significantly decreasing the volume of the loader.

(45) Depressing or otherwise actuating the release mechanism **940** releases the body portions **902**, **903** by disengaging the latch **916** from the catch **917** so that they could be separated as shown by the arrow A in FIG. **20**. The body portions remain connected by the hinge at the front **932** of the body **901**.

(46) In the illustrated embodiment of FIGS. **9-13**, the release mechanism **940** extends from a proximal end **941** near the hinge **905** to a distal end **202** toward the middle of the loader body **901**. As shown, the release mechanism **940** extends along and near the lower front wall **914**, but may alternatively be positioned along the sides or rear of the body **901** or portions(s) thereof.

(47) Because the player may not be able to see the release mechanism **940** while operating it, various ergonomic features may be included. In the illustrated embodiment, the release mechanism **940** is an oversized paddle-like lever with a width that increases to complement the shape of the loader body **901**. The release mechanism **940** may also include a textured surface **204** and/or a raised rib **205** to guide operation.

(48) As best shown in FIG. **13**, pressing the paddle-like lever (or other actuation) of release mechanism **940** causes pivoting motion about a pivot **946**, which rotates a flange **948** away from the catch **917**. The flange is operatively connected to a bowden cable **950** or other mechanical linkage to release the latch **916** from the catch **917**.

(49) The bowden cable can be routed around a side or the lower wall **911**, **914** of the body **901** of the projectile loader **900**. The latch **916** moves linearly in a direction B in response to the bowden cable **950** and is biased to the closed position by a return spring **920**. Alternatively, the release mechanism can actuate a solenoid to release the latch **916** from the catch **917**. Various combinations of release mechanisms could also be used.

(50) The release mechanism **940** may have various other additions or modifications. For example, the release mechanism **940** may provide an audible “click” or similar feedback when successfully actuated. The release mechanism **940** may require a two-step actuation motion, or the actuation of two separate buttons, so as to prevent accidental actuation. A spring, such as a torsion spring **952** or a compression spring, may bias the upper body portion **902** to either the open position or the closed position, as desired. In one embodiment, the torsion spring **952** is mounted coaxially with the hinge **905** to both the first and second portions **902**, **903** to bias toward the open position.

(51) Another feed mechanism **920** applicable to the projectile loader **900** of the present invention is shown in FIG. **14**. The feed mechanism **920** includes one or more of the following: a battery holder **921**, manual drive assist knob **922**, motor **923** with battery harness and battery, drive shaft **925**, drive carrier (feeder) **926**, circuit board with microprocessor **927**, floor **928**, back bone, or frame **929** including an outfeed tube or feed neck **930**. All of the components are housed on the frame **929** to form an integral feed mechanism assembly **920** which may be removed as a unit. This feed mechanism **920** is applicable to the PROPHECY® or Z2™ brand projectile loaders, among others.

(52) In any of the embodiments discussed above, the first portion **102/902** and second portion **103/903** may be additionally secured to each other via any selected engagement means, such as, by way of example and not by way of limitation, a friction fit, a snapping engagement, complementary interfitted structure, or any other latching mechanism. One portion of the body may have an extension snappingly received in an opening in the other portion of the body. The first portion **902** and second portion **903** may be held together with a removable pin such as a cotter-type pin or a spring biased pin that projects through holes formed in each the portions. Such structure(s) may be positioned at one or multiple areas where the first portion **102/902** engages the second portion **103/903**. In other words, the first portion **902** and second portion **903** may have releasable attachments at a plurality of locations along the meeting or facing edges of the body portions. These structures and attachments may be only engaged by a friction or force fit, such that actuation of the release mechanism **140/940** is still sufficient to open the loader body **101/901**.

(53) The arrangement of the present invention, having an internally-accessed release mechanism allowing for a simplified exterior, provides advantages over the prior art. The internal components

can be readily and fully accessed when the first and second portions are separated. The hinge allows the first and second portions to stay attached when the body is opened. The simplified exterior provides a streamlined look while protecting the release mechanism from dirt, debris, or accidental actuation.

(54) It is understood that this invention is not limited to the particular embodiments disclosed, but is intended to cover all modifications and combinations which are within the spirit and scope of the invention as described herein and/or defined by the appended claims, the above description, and/or shown in the attached drawings.

Claims

1. A projectile loader, comprising: a body having an upper portion and defining an interior area for housing internal components, the body comprising an opening for loading projectiles into the projectile loader; the body being split into a first portion and a second portion; the first portion attached to the second portion, the first portion and second portion moveable relative to each other between an open position and a closed position; and a release mechanism configured to selectively permit detaching the first portion and the second portion, wherein an actuator of the release mechanism is located completely within the interior area of the body, and wherein the actuator of the release mechanism is only accessible via the opening.
2. The projectile loader of claim 1, wherein the projectile loader has an outer surface that is free from external features other than a single recess for a power or operation button.
3. The projectile loader of claim 1, wherein the projectile loader has a front and a rear opposite the front and wherein the release mechanism and the opening are both located proximate to the rear of the projectile loader.
4. The projectile loader of claim 1, wherein the projectile loader has a front and a rear opposite the front and the opening is located proximate to the rear of the projectile loader.
5. The projectile loader of claim 4, wherein the projectile loader has a front and a rear opposite the front and wherein the release mechanism is located proximate to the rear.
6. The projectile loader of claim 1, wherein the release mechanism is configured for one-handed actuation.
7. The projectile loader of claim 1, wherein the first portion and the second portion have attachment surfaces that are essentially a same size.
8. The projectile loader of claim 1, further comprising a moveable feeder housed within the interior area of the projectile loader body.
9. The projectile loader of claim 1, further comprising a feed mechanism comprising a moveable feeder and a motor housed within the interior area of the projectile loader body, the feed mechanism configured to be at least partially removable from the projectile loader body when the first portion and second portion of the body are in the open position.
10. The projectile loader of claim 1, further comprising a hinge connecting the first portion and the second portion, wherein the hinge is spring-biased to an open position.
11. The projectile loader of claim 1, further comprising a hinge connecting the first portion and the second portion, wherein the hinge is spring-biased to a closed position.
12. The projectile loader of claim 1, wherein the body is split generally vertically along a length of the body such that the first and second portions are respectively a first side portion and a second side portion.
13. A projectile loader, comprising: a body having an upper portion and defining an interior area for housing internal components, the body comprising an opening for loading projectiles into the projectile loader; the body being split into a first portion and a second portion; the first portion attached to the second portion, the first portion and second portion moveable relative to each other between an open position and a closed position; moveable feeder housed within the interior area of

the projectile loader body; and a release mechanism configured to selectively permit detaching the first portion and the second portion, wherein an actuator of the release mechanism is located completely within the interior area of the body, and wherein the actuator of the release mechanism is only accessible via the opening.

14. The projectile loader of claim 13, wherein the projectile loader has an outer surface that is free from external features other than a single recess for a power or operation button.

15. The projectile loader of claim 13, wherein the projectile loader has a front and a rear opposite the front and wherein the release mechanism and the opening are both located proximate to a rear of the projectile loader.

16. The projectile loader of claim 13, the projectile loader has a front and a rear opposite the front and wherein the release mechanism is located proximate to the rear.

17. The projectile loader of claim 13, wherein the projectile loader has a front and a rear opposite the front and wherein the release mechanism is located proximate to the rear.

18. The projectile loader of claim 13, wherein the release mechanism is configured for one-handed actuation.

19. The projectile loader of claim 13, wherein the first portion and the second portion have attachment surfaces that are essentially a same size.

20. The projectile loader of claim 13, further comprising a feed mechanism housed within the interior area of the projectile loader, the feed mechanism comprising the moveable feeder and a motor for actuating the moveable feeder, wherein the feed mechanism is configured to be at least partially removable from the projectile loader body when the first portion and the second portion of the body are in the open position.
